Lesson 2: Variables, Data Types, and Choosing Graphs

Unit

Working Like a Physicist: Collecting, Processing, and Presenting Data

Lesson focus

Students learn the basic language of experiments and decide when to use a line graph or a bar chart.

Learning objectives

By the end of the lesson, students should be able to:

1. Identify independent, dependent, and control variables.
2. Distinguish between continuous and categoric data.
3. Decide whether data should be shown using a line graph or a bar chart.
4. Explain their graph choice using the words continuous or categoric.

Key vocabulary

Keyword	Student-friendly definition
Independent variable	The thing you change
Dependent variable	The thing you measure
Control variable	Something you keep the same
Continuous data	Number data that can have many possible values
Categoric data	Data sorted into groups or types
Line graph	A graph used when the independent variable is continuous
Bar chart	A graph used when the independent variable is categoric

Equipment and resources

- Mini whiteboards or exercise books
- Projected examples of investigation questions
- Optional: toy car, ramp, different surfaces
- Optional: worksheet with investigation sorting activity

Starter: What are we changing?

Give students four investigation questions:

1. How does ramp height affect trolley speed?
2. Which material is the best thermal insulator?
3. How does the number of elastic bands affect launch distance?
4. Which drink contains the most sugar?

For each one, students identify:

- what is changed
- what is measured
- what must be kept the same

Introduce the three main variable types:

Term	Meaning
Independent variable	The variable you change
Dependent variable	The variable you measure
Control variable	A variable you keep the same

Main teaching

Introduce continuous and categoric data.

Type of data	Meaning	Example	Best graph
Continuous data	Numerical data that can take many values	Length, time, temperature, mass	Line graph
Categoric data	Data in groups or names	Material type, color, brand	Bar chart

Key rule:

> *Use a line graph when the independent variable is continuous.*
> *Use a bar chart when the independent variable is categoric.*

Activity 1: Sort the investigations

Students sort the following investigation titles into line graph or bar chart.

Investigation	Graph type
How does temperature affect dissolving time?	Line graph
Which surface gives the most friction?	Bar chart
How does mass affect stretching of a spring?	Line graph
Which paper towel absorbs the most water?	Bar chart
How does the length of a pendulum affect its time period?	Line graph
Which material is the best sound absorber?	Bar chart

Students should justify each choice using continuous or categoric.

Optional mini-demonstration

Demonstration title

Which surface slows a toy car most?

Method

Roll a toy car down the same ramp onto different surfaces, for example:

- bench
- carpet
- rubber mat
- sandpaper

Students identify:

- independent variable: surface type
- dependent variable: distance travelled
- control variables: same car, same ramp height, same starting point

They should decide that this would need a bar chart because surface type is categoric.

Check for understanding

Ask students:

1. Is surface type continuous or categoric?
2. Is temperature continuous or categoric?
3. Is material type continuous or categoric?
4. Is length continuous or categoric?
5. Which graph should we use when the independent variable is continuous?
6. Which graph should we use when the independent variable is categoric?

Plenary: Exit ticket

Students answer:

1. What is the independent variable?
2. What is the dependent variable?
3. When should you use a line graph?
4. When should you use a bar chart?

Possible prep

Students classify 10 investigation questions as requiring either a line graph or a bar chart. For each one, they must explain whether the independent variable is continuous or categoric.

Teacher notes

Common misconception:

Students may think that any numerical dependent variable means they should use a line graph. Emphasize that the graph choice depends mainly on the independent variable.